Reg No.

ne : 3.00 hrs. Max. Marks : 70

Quarterly Examination - 2025

BIOLOGY

BIO - BOTANY

Section – A

Answer all the questions. 8x1=8

Coelinhiza is found in _____ a) Paddy b) Bean c) Pea d) Tridax

Which of the following represent Megagamelophyte.

a) Ovule b) Embryo sac c) Nucelluss d) Endosperm

The time duration for sterilization process by using autoclave is _____ minutes and the temperature is _____. a) 10 to 30 minutes and 125°C b) 15 to 30 minutes and 121°C

c) 15 to 20 minutes and 125°C d) 10 to 20 minutes and 121°C

Virus free plants are developed from _____

a) Organ culture b) Meristem culture c) Protoplast culture d) Cell suspension culture

Which of the following represent Inhibition gene F_2-Pheno-typic ratio-

a) 9:3:4 b) 9:7 c) 13:3 d) 15:1

Which one of the following is not correct statement.

a) Ti plasmid causes the bunchy top disease b) Multiple cloning site is known as Polylinker

c) Non viral method transfection of Nucleic acid in all

d) Poly lactic acid is a kind of biodegradable and bioactive thermoplastic.

Due to incomplete linkage in maize, the ratio of parental and recombinants are

a) 50:50 b) 7:1:1:7 c) 96:4:3:6 d) 1:7:7:1

Who reported a lethal gene _____ a) Linnaeus b) Carl Correns c) E.Baur d) P.Maheswari

SECTION - B

Answer any four questions. 4x2=8

What is pollen kit?

Give the name of the scientists who rediscovered Mendalism.

What is bioremediation?

Draw a diagram of Aneuploidy – tetrasomy

What is Back Cross?

Define – Cybrid

SECTION-C

Answer any three questions. (Question No.19 is compulsory) 3x3=9

List out the function of Tapetum.

What is gene mapping? Write its uses. (any 2)

Give an account on cryopreservation.

What is meant by cytoplasmic inheritance?

Mention the name of man-made cereal. Draw its flow chart.

SECTION-D

Answer all questions. 2x5=10

20. A. Explain the structure of an ovule with a suitable diagram. **(OR)**

B. Bring out the inheritance of chloroplast gene with an example.

21. A. What are restriction enzyme? Mention their type with role in biotechnology. **(OR)**

B. Explain the basic concepts involved in plant tissue culture.

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BIO - ZOOLOGY

Section-A

I. Answer all the questions. 8x

1. Spread of cancerous cells to distant sites is termed as
   a) Metastasis b) Oncogenes c) proto-oncogenes d) malignant neoplasm
2. Exo-erythrocytic schizogony of plasmodium take place in
   a) RBC b) Leucocytes c) Stomach d) Liver
3. The golden age of reptiles was
   a) Mesozoic era b) Cenzoic era c) Paleozoic era d) Proterozoic era
4. Which of the following was contribution of Hugo de virus?
   a) Theory of Mutation b) Theory of natural selection c) Theory of Inheritance of acquired characters d) Germplasm theory
5. A mRNA molecule is produced by
   a) Replication b) Transcription c) Duplication d) Translation
6. ZW-ZZ system of sex determination occurs in
   a) fishes b) reptiles c) birds d) all of these
7. Colostrum is rich in a) IgE b) IgA c) IgD d) IgM
8. Which one is correct match.
   a) Autogamy – Trichonympha b) Merogmy – Monocystis
   c) Exogamy – Human d) Isogamy – Higher Invertebrates

SECTION - B

II. Answer any four questions. 4x2=

9. What is Inhibin?
10. Define - Plasmotomy.
11. Differentiate between convergent evolution divergent evolution.
12. What is coacervates?
13. What is lyonisation?
14. What is genetic code?

SECTION - C

III. Answer any three questions. (Question No.19 is compulsory) 3x3=

15. Why the human genome project is called a mega project?
16. Give the salient features of Mutation theory.
17. Draw a labelled sketch of a Spermatozoa.
18. Define – POCSO Act.
19. Differentiate between the following:
    External fertilization and Internal fertilization.

SECTION - D

IV. Answer all questions: 2x5=10

20. A) Give a schematic representation of spermotogenesis and Oogenesis in human.
    (OR)
    B) Methodologies of Human Gnome project.
21. A) Explain the evolutionary path of man.
    (OR)
    B) Explain the structure of immunoglobulin with suitable diagram.

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